MPG No. 7.4.23 $m'm'2$ [Type III, orthorhombic] [T tensor]

* Rank 0 tensor. * Rank 1 tensor. * Rank 2 tensor (s).

$$\begin{bmatrix} 0 & T_{xy} & 0 \\ T_{xy} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$T_{xy} = \mathbb{T}^{(1)}_{xy}$$

* Rank 2 tensor (a).

$$\begin{bmatrix} 0 & T_{xy} & 0 \\ -T_{xy} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$T_{xy} = \mathbb{M}^{(1)}_{z}$$

* Rank 3 tensor (ss).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ T_{yzx} & 0 & 0 \\ 0 & T_{yzx} & 0 \\ 0 & 0 & T_{yzx} \end{bmatrix}$$

$$T_{yzx} = \mathbb{T}^{(1)}_{xyz}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ T_{yzx} & 0 & 0 \\ 0 & T_{zxy} & 0 \\ 0 & 0 & -T_{yzx} - T_{zxy} \end{bmatrix}$$

$$T_{yzx} = 3\mathbb{M}^{(1)}_{u} + \mathbb{M}^{(1)}_{v}$$
$$T_{zxy} = -3\mathbb{M}^{(1)}_{u} + \mathbb{M}^{(1)}_{v}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} T_{yzx} & 0 & 0 \\ 0 & T_{zxy} & 0 \\ 0 & 0 & T_{xyz} \end{bmatrix}$$

$$T_{yzx} = \mathbb{M}^{(1)}_{0} - \mathbb{M}^{(2)}_{u} + \mathbb{M}^{(2)}_{v}$$
$$T_{zxy} = \mathbb{M}^{(1)}_{0} - \mathbb{M}^{(2)}_{u} - \mathbb{M}^{(2)}_{v}$$
$$T_{xyz} = \mathbb{M}^{(1)}_{0} + 2\mathbb{M}^{(2)}_{u}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & T_{xxxy} \\ 0 & 0 & 0 & 0 & 0 & T_{yyxy} \\ 0 & 0 & 0 & 0 & 0 & T_{zzxy} \\ 0 & 0 & 0 & 0 & T_{yzzx} & 0 \\ 0 & 0 & 0 & T_{yzzx} & 0 & 0 \\ T_{xxxy} & T_{yyxy} & T_{zzxy} & 0 & 0 & 0 \end{bmatrix}$$

$$T_{xxxy} = \mathbb{T}^{\alpha(1)}_{4z} - \mathbb{T}^{\beta(1)}_{4z} + \mathbb{T}^{(1)}_{xy} + 2\mathbb{T}^{(2)}_{xy}$$
$$T_{yyxy} = -\mathbb{T}^{\alpha(1)}_{4z} - \mathbb{T}^{\beta(1)}_{4z} + \mathbb{T}^{(1)}_{xy} + 2\mathbb{T}^{(2)}_{xy}$$
$$T_{zzxy} = 2\mathbb{T}^{\beta(1)}_{4z} + \mathbb{T}^{(1)}_{xy}$$
$$T_{yzzx} = 2\mathbb{T}^{\beta(1)}_{4z} + \mathbb{T}^{(2)}_{xy}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & T_{xxxy} \\ 0 & 0 & 0 & 0 & 0 & T_{yyxy} \\ 0 & 0 & 0 & 0 & 0 & T_{zzxy} \\ 0 & 0 & 0 & 0 & T_{yzzx} & 0 \\ 0 & 0 & 0 & -T_{yzzx} & 0 & 0 \\ -T_{xxxy} & -T_{yyxy} & -T_{zzxy} & 0 & 0 & 0 \end{bmatrix}$$

$$T_{xxxy} = 2\mathbb{M}^{(1)}_{z} - 2\mathbb{M}^{\alpha(1)}_{z} + 2\mathbb{M}^{\beta(1)}_{z} + \mathbb{T}^{(3)}_{xy}$$
$$T_{yyxy} = -2\mathbb{M}^{(1)}_{z} + 2\mathbb{M}^{\alpha(1)}_{z} + 2\mathbb{M}^{\beta(1)}_{z} + \mathbb{T}^{(3)}_{xy}$$
$$T_{zzxy} = -4\mathbb{M}^{\beta(1)}_{z} + \mathbb{T}^{(3)}_{xy}$$
$$T_{yzzx} = -\mathbb{M}^{(1)}_{z} - 4\mathbb{M}^{\alpha(1)}_{z}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} 0 & T_{yzzx} & 0 \\ T_{yzzx} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$T_{yzzx} = 2\mathbb{T}^{(4)}_{xy}$$

* Rank 4 tensor (aaa).

$$\begin{bmatrix} 0 & T_{yzzx} & 0 \\ -T_{yzzx} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$T_{yzzx} = -\mathbb{M}^{(2)}_{z}$$

* Rank 4 tensor (sa).

$$\begin{bmatrix} 0 & 0 & T_{xxxy} \\ 0 & 0 & T_{yyxy} \\ 0 & 0 & T_{zzxy} \\ 0 & T_{yzzx} & 0 \\ T_{zxyz} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$T_{xxxy} = \mathbb{M}^{(3)}_{z} - \mathbb{M}^{\alpha(2)}_{z} + \mathbb{M}^{\beta(2)}_{z} - 2\mathbb{T}^{(5)}_{xy}$$
$$T_{yyxy} = \mathbb{M}^{(3)}_{z} - \mathbb{M}^{\alpha(2)}_{z} - \mathbb{M}^{\beta(2)}_{z} + 2\mathbb{T}^{(5)}_{xy}$$
$$T_{zzxy} = \mathbb{M}^{(3)}_{z} + 2\mathbb{M}^{(4)}_{z} + 2\mathbb{M}^{\alpha(2)}_{z}$$
$$T_{yzzx} = \mathbb{M}^{(4)}_{z} - \mathbb{M}^{\alpha(2)}_{z} - \mathbb{M}^{\beta(2)}_{z} - \mathbb{T}^{(5)}_{xy}$$
$$T_{zxyz} = \mathbb{M}^{(4)}_{z} - \mathbb{M}^{\alpha(2)}_{z} + \mathbb{M}^{\beta(2)}_{z} + \mathbb{T}^{(5)}_{xy}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & T_{yzzx} & 0 \\ 0 & 0 & 0 & T_{zxyz} & 0 & 0 \\ T_{xyxx} & T_{xyyy} & T_{xyzz} & 0 & 0 & 0 \end{bmatrix}$$

$$T_{yzzx} = \mathbb{M}_z^{(6)} - \mathbb{M}_z^{\alpha(3)} + \mathbb{M}_z^{\beta(3)} + \mathbb{T}_{xy}^{(6)}$$
$$T_{zxyz} = \mathbb{M}_z^{(6)} - \mathbb{M}_z^{\alpha(3)} - \mathbb{M}_z^{\beta(3)} - \mathbb{T}_{xy}^{(6)}$$
$$T_{xyxx} = \mathbb{M}_z^{(5)} - \mathbb{M}_z^{\alpha(3)} + \mathbb{M}_z^{\beta(3)} - 2\mathbb{T}_{xy}^{(6)}$$
$$T_{xyyy} = \mathbb{M}_z^{(5)} - \mathbb{M}_z^{\alpha(3)} - \mathbb{M}_z^{\beta(3)} + 2\mathbb{T}_{xy}^{(6)}$$
$$T_{xyzz} = \mathbb{M}_z^{(5)} + 2\mathbb{M}_z^{(6)} + 2\mathbb{M}_z^{\alpha(3)}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & T_{xxxy} & 0 & 0 & T_{xxyx} \\ 0 & 0 & 0 & 0 & 0 & T_{yyxy} & 0 & 0 & T_{yyyx} \\ 0 & 0 & 0 & 0 & 0 & T_{zzxy} & 0 & 0 & T_{zzyx} \\ 0 & 0 & 0 & 0 & T_{yzzx} & 0 & 0 & T_{yzxz} & 0 \\ 0 & 0 & 0 & T_{zxyz} & 0 & 0 & T_{zxzy} & 0 & 0 \\ T_{xyxx} & T_{xyyy} & T_{xyzz} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$T_{xxxy} = 2\mathbb{M}_z^{(1)} + \mathbb{M}_z^{(3)} - 2\mathbb{M}_z^{\alpha(1)} - \mathbb{M}_z^{\alpha(2)} + 2\mathbb{M}_z^{\beta(1)} + \mathbb{M}_z^{\beta(2)} + \mathbb{T}_{4z}^{\alpha(1)} - \mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} + \mathbb{T}_{xy}^{(3)} - 2\mathbb{T}_{xy}^{(5)}$$
$$T_{xxyx} = 2\mathbb{M}_z^{(1)} - \mathbb{M}_z^{(3)} - 2\mathbb{M}_z^{\alpha(1)} + \mathbb{M}_z^{\alpha(2)} + 2\mathbb{M}_z^{\beta(1)} - \mathbb{M}_z^{\beta(2)} + \mathbb{T}_{4z}^{\alpha(1)} - \mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} + \mathbb{T}_{xy}^{(3)} + 2\mathbb{T}_{xy}^{(5)}$$
$$T_{yyxy} = -2\mathbb{M}_z^{(1)} + \mathbb{M}_z^{(3)} + 2\mathbb{M}_z^{\alpha(1)} - \mathbb{M}_z^{\alpha(2)} + 2\mathbb{M}_z^{\beta(1)} - \mathbb{M}_z^{\beta(2)} - \mathbb{T}_{4z}^{\alpha(1)} - \mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} + \mathbb{T}_{xy}^{(3)} + 2\mathbb{T}_{xy}^{(5)}$$
$$T_{yyyx} = -2\mathbb{M}_z^{(1)} - \mathbb{M}_z^{(3)} + 2\mathbb{M}_z^{\alpha(1)} + \mathbb{M}_z^{\alpha(2)} + 2\mathbb{M}_z^{\beta(1)} + \mathbb{M}_z^{\beta(2)} - \mathbb{T}_{4z}^{\alpha(1)} - \mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} + \mathbb{T}_{xy}^{(3)} - 2\mathbb{T}_{xy}^{(5)}$$
$$T_{zzxy} = \mathbb{M}_z^{(3)} + 2\mathbb{M}_z^{(4)} + 2\mathbb{M}_z^{\alpha(2)} - 4\mathbb{M}_z^{\beta(1)} + 2\mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} + \mathbb{T}_{xy}^{(3)}$$
$$T_{zzyx} = -\mathbb{M}_z^{(3)} - 2\mathbb{M}_z^{(4)} - 2\mathbb{M}_z^{\alpha(2)} - 4\mathbb{M}_z^{\beta(1)} + 2\mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} + \mathbb{T}_{xy}^{(3)}$$
$$T_{yzzx} = -\mathbb{M}_z^{(1)} + \mathbb{M}_z^{(4)} - 4\mathbb{M}_z^{\alpha(1)} - \mathbb{M}_z^{\alpha(2)} - \mathbb{M}_z^{\beta(2)} + 2\mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(2)} - \mathbb{T}_{xy}^{(5)}$$
$$T_{yzxz} = -\mathbb{M}_z^{(1)} - \mathbb{M}_z^{(4)} - 4\mathbb{M}_z^{\alpha(1)} + \mathbb{M}_z^{\alpha(2)} + \mathbb{M}_z^{\beta(2)} + 2\mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(2)} + \mathbb{T}_{xy}^{(5)}$$
$$T_{zxyz} = \mathbb{M}_z^{(1)} + \mathbb{M}_z^{(4)} + 4\mathbb{M}_z^{\alpha(1)} - \mathbb{M}_z^{\alpha(2)} + \mathbb{M}_z^{\beta(2)} + 2\mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(2)} + \mathbb{T}_{xy}^{(5)}$$
$$T_{zxzy} = \mathbb{M}_z^{(1)} - \mathbb{M}_z^{(4)} + 4\mathbb{M}_z^{\alpha(1)} + \mathbb{M}_z^{\alpha(2)} - \mathbb{M}_z^{\beta(2)} + 2\mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(2)} - \mathbb{T}_{xy}^{(5)}$$
$$T_{xyxx} = -2\mathbb{M}_z^{(1)} + 2\mathbb{M}_z^{\alpha(1)} - 2\mathbb{M}_z^{\beta(1)} + \mathbb{T}_{4z}^{\alpha(1)} - \mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} - \mathbb{T}_{xy}^{(3)}$$
$$T_{xyyy} = 2\mathbb{M}_z^{(1)} - 2\mathbb{M}_z^{\alpha(1)} - 2\mathbb{M}_z^{\beta(1)} - \mathbb{T}_{4z}^{\alpha(1)} - \mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} - \mathbb{T}_{xy}^{(3)}$$
$$T_{xyzz} = 4\mathbb{M}_z^{\beta(1)} + 2\mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} - \mathbb{T}_{xy}^{(3)}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & T_{yzzx} & 0 & 0 & T_{yzxz} & 0 \\ 0 & 0 & 0 & T_{zxyz} & 0 & 0 & T_{zxzy} & 0 & 0 \\ T_{xyxx} & T_{xyyy} & T_{xyzz} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$T_{yzzx} = -\mathbb{M}_z^{(2)} + \mathbb{M}_z^{(6)} - \mathbb{M}_z^{\alpha(3)} + \mathbb{M}_z^{\beta(3)} + 2\mathbb{T}_{xy}^{(4)} + \mathbb{T}_{xy}^{(6)}$$
$$T_{yzxz} = \mathbb{M}_z^{(2)} + \mathbb{M}_z^{(6)} - \mathbb{M}_z^{\alpha(3)} + \mathbb{M}_z^{\beta(3)} - 2\mathbb{T}_{xy}^{(4)} + \mathbb{T}_{xy}^{(6)}$$
$$T_{zxyz} = \mathbb{M}_z^{(2)} + \mathbb{M}_z^{(6)} - \mathbb{M}_z^{\alpha(3)} - \mathbb{M}_z^{\beta(3)} + 2\mathbb{T}_{xy}^{(4)} - \mathbb{T}_{xy}^{(6)}$$
$$T_{zxzy} = -\mathbb{M}_z^{(2)} + \mathbb{M}_z^{(6)} - \mathbb{M}_z^{\alpha(3)} - \mathbb{M}_z^{\beta(3)} - 2\mathbb{T}_{xy}^{(4)} - \mathbb{T}_{xy}^{(6)}$$
$$T_{xyxx} = \mathbb{M}_z^{(5)} - \mathbb{M}_z^{\alpha(3)} + \mathbb{M}_z^{\beta(3)} - 2\mathbb{T}_{xy}^{(6)}$$
$$T_{xyyy} = \mathbb{M}_z^{(5)} - \mathbb{M}_z^{\alpha(3)} - \mathbb{M}_z^{\beta(3)} + 2\mathbb{T}_{xy}^{(6)}$$
$$T_{xyzz} = \mathbb{M}_z^{(5)} + 2\mathbb{M}_z^{(6)} + 2\mathbb{M}_z^{\alpha(3)}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} 0 & T_{xxxy} & 0 \\ T_{yyyx} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & T_{zzxy} & 0 \\ T_{xxyx} & 0 & 0 \\ T_{yzzx} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & T_{xyyy} & 0 \\ 0 & 0 & \frac{T_{xxxy}}{2} - \frac{T_{xxyx}}{2} + T_{xyyy} - T_{yyyx} + T_{zzxy} \end{bmatrix}$$

$$T_{xxxy} = \mathbb{T}^{\alpha(1)}_{4z} - \mathbb{T}^{\beta(1)}_{4z} + \mathbb{T}^{(1)}_{xy} + 2\mathbb{T}^{(2)}_{xy} + \mathbb{T}^{(3)}_{xy} - 2\mathbb{T}^{(5)}_{xy}$$
$$T_{yyyx} = -\mathbb{T}^{\alpha(1)}_{4z} - \mathbb{T}^{\beta(1)}_{4z} + \mathbb{T}^{(1)}_{xy} + 2\mathbb{T}^{(2)}_{xy} + \mathbb{T}^{(3)}_{xy} - 2\mathbb{T}^{(5)}_{xy}$$
$$T_{zzxy} = 2\mathbb{T}^{\beta(1)}_{4z} + \mathbb{T}^{(1)}_{xy} + \mathbb{T}^{(3)}_{xy}$$
$$T_{xxyx} = \mathbb{T}^{\alpha(1)}_{4z} - \mathbb{T}^{\beta(1)}_{4z} + \mathbb{T}^{(1)}_{xy} + 2\mathbb{T}^{(2)}_{xy} + \mathbb{T}^{(3)}_{xy} + 2\mathbb{T}^{(5)}_{xy}$$
$$T_{yzzx} = 2\mathbb{T}^{\beta(1)}_{4z} + \mathbb{T}^{(2)}_{xy} - \mathbb{T}^{(5)}_{xy}$$
$$T_{xyyy} = -\mathbb{T}^{\alpha(1)}_{4z} - \mathbb{T}^{\beta(1)}_{4z} + \mathbb{T}^{(1)}_{xy} + 2\mathbb{T}^{(2)}_{xy} - \mathbb{T}^{(3)}_{xy}$$